Data Quality Report

Team ID	MCA
Project Title	Smartphone Usage & Digital Well-Being
Maximum Marks	3 Marks

The Data Quality Report summarizes data quality issues identified during the data collection and preprocessing phase for the Smartphone_Usage.csv dataset. This dataset contains 7,500 user records across 16 behavioral and demographic attributes, focusing on mapping digital addiction trends. The template outlines structural inconsistencies, missing metrics, and anomalies alongside technical resolution plans to ensure accurate downstream statistical analytics.

Data Quality Issue Log & Resolution Matrix

Data Source	Data Quality Issue	Severity	Resolution Plan
Smartphone_Usage.csv (Demographics)	Missing fields discovered within the user 'Age' attribute, which acts as a foundational segmentation filter.	**Moderate**	Utilized Power Query's conditional column structures to isolate missing entries, then imputed missing demographic fields using the localized median value across matching gender strata.
Smartphone_Usage.csv (System Logs)	Duplicate session IDs and redundant transaction logs generated by multi-device sync processes, causing sample inflation.	**High**	Applied a strict deduplication logic string via Power Query ('Remove Duplicates') based on unique user ID timestamps to preserve unique baseline records.
Smartphone_Usage.csv (Metrics)	Inconsistent numeric scale constraints where weekend screen time values	**High**	Configured structural data validation check limits. Built a technical boundary replacement rule replacing

	exceeded expected physiological thresholds or logged negative figures.		out-of-bound anomalies with max logical boundaries or dataset averages.
Smartphone_Usage.csv (App Opens)	Data type anomalies where raw count data columns for system notification checks and app openings were parsed as text strings rather than numbers.	**Low**	Executed an explicit explicit data type conversion step within the Power Query transformation engine, strictly transforming columns into 64-bit Integer formatting.
Smartphone_Usage.csv (Target Variable)	Class imbalance skew within the target binary field 'Addicted Label' (1 = Addicted, 0 = Not Addicted), creating potential bias risks.	**Moderate**	Established stratified random sampling controls during data modeling slices to guarantee analytical integrity and metric equity between the classification sets.